

# ITERATED SUBGROUP-NORMALIZER DYNAMICS IN FINITE DIHEDRAL GROUPS

ANONYMOUS

ABSTRACT. Let  $D_{2n} = \langle r, s \mid r^n = s^2 = 1, srs = r^{-1} \rangle$ , and iterate  $H \mapsto N_{D_{2n}}(H)$  on the set of all subgroups. The subgroup coordinates and the one-step odd/even normalizer rule are classical and are used here with full credit to their sources. Writing  $n = 2^a m$  with  $m$  odd, we show that the remaining dynamics consists of  $\sigma(m)$  full binary inverse trees of height  $a$ , together with all  $\tau(n)$  rotation subgroups feeding one distinguished fixed root. We derive every positive-time image and every target fibre. We then prove that the unlabelled functional graph is classified exactly by  $(a, \sigma(m), \tau(n))$ . In particular, the parameters 33 and 35 give conjugate graphs, and so do all common power-of-two lifts. Symbolic proofs are paired with deterministic literal-normalizer replay.

## 1. SCOPE, OWNED INPUTS, AND THEOREM

The subgroup structure of finite dihedral groups is classical [1, 2]. The normalizer of a reflection-containing dihedral subgroup is likewise governed by a direct odd/even rule; it appears in Frenkel [3] and in complete subgroup cases in Shelash et al. [4]. Those results settle the carrier and the one-step map. They receive no contribution credit below. Our subject is what repeated application does to *every* subgroup, including its full inverse forest, all-time fibres, and the information about  $n$  retained by the resulting unlabelled graph.

Put

$$G_n = D_{2n} = \langle r, s \mid r^n = s^2 = 1, srs = r^{-1} \rangle, \quad n \geq 3.$$

For  $d \mid n$  and  $0 \leq j < d$ , write

$$R_d = \langle r^d \rangle, \quad H_{d,j} = \langle r^d, r^j s \rangle.$$

The complete subgroup carrier is

$$(1) \quad \text{Sub}(G_n) = \{R_d : d \mid n\} \sqcup \{H_{d,j} : d \mid n, 0 \leq j < d\},$$

of size  $\tau(n) + \sigma(n)$ . Let  $\mathcal{N}_n(H) = N_{G_n}(H)$ , and write  $n = 2^a m$  with  $m$  odd.

**Theorem 1** (Full iterated normalizer graph). *For every  $n \geq 3$ , the map  $\mathcal{N}_n$  on  $\text{Sub}(G_n)$  has the following properties.*

(i) *The owned one-step rule and its iterates are*

$$(2) \quad \mathcal{N}_n(R_d) = H_{1,0}, \quad \mathcal{N}_n(H_{d,j}) = H_{d/\gcd(d,2), j \bmod d/\gcd(d,2)},$$

$$(3) \quad \mathcal{N}_n^t(H_{2^k e, j}) = H_{2^{\max(k-t,0)} e, j \bmod 2^{\max(k-t,0)} e},$$

for  $t \geq 0$ ,  $0 \leq k \leq a$ ,  $e \mid m$ , and  $0 \leq j < 2^k e$ .

(ii) *The dihedral states form  $\sigma(m)$  full binary inverse trees of height  $a$ , rooted at the fixed states  $H_{e,j}$  with  $e \mid m$ ,  $0 \leq j < e$ . Every  $R_d$  is an extra source of depth one entering only  $H_{1,0} = G_n$ . Hence*

$$\text{depth}(R_d) = 1, \quad \text{depth}(H_{d,j}) = v_2(d),$$

and the depth polynomial is

$$(4) \quad \mathcal{D}_n(z) = \sigma(m) + \tau(n)z + \sigma(m) \sum_{k=1}^a 2^k z^k.$$

All cycles are fixed points, and  $\#\text{Fix}(\mathcal{N}_n^t) = \sigma(m)$  for  $t \geq 1$ .

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(iii) For every  $t \geq 1$ ,

$$(5) \quad |\text{im } \mathcal{N}_n^t| = \sigma(m)(2^{\max(a-t,0)+1} - 1).$$

No rotation target has a positive-time preimage. For every  $t \geq 1$  and every positive-level carrier target  $H_{2^k e, j}$  with  $1 \leq k \leq a$ ,  $e \mid m$ , and  $0 \leq j < 2^k e$ ,

$$(6) \quad \#(\mathcal{N}_n^t)^{-1}(H_{2^k e, j}) = \begin{cases} 2^t, & k+t \leq a, \\ 0, & k+t > a, \end{cases}$$

while a fixed root has fibre

$$(7) \quad \#(\mathcal{N}_n^t)^{-1}(H_{e, j}) = 2^{\min(t, a)+1} - 1 + \tau(n) \mathbf{1}_{(e, j) = (1, 0)}.$$

(iv) Two such unlabelled directed functional graphs are isomorphic exactly when

$$(8) \quad (v_2(n), \sigma(\text{odd}(n)), \tau(n)) = (v_2(q), \sigma(\text{odd}(q)), \tau(q)).$$

In particular, the graphs for  $n = 33$  and  $n = 35$  are conjugate, although  $|G_{33}| \neq |G_{35}|$ ; the same holds for  $(2^b 33, 2^b 35)$  for every  $b \geq 0$ .

Proof dependency graph. The theorem is organized so the owner-covered bridge is visibly separated from the dynamical deductions:

$$(9) \quad \begin{array}{c} \boxed{\text{subgroup coordinates} + \text{one-step parity rule}} \\ \Downarrow \\ \boxed{t\text{-fold halving (3)}} \\ \Downarrow \\ \text{binary forest} \implies \{\mathcal{D}_n, \text{im } \mathcal{N}_n^t, \text{target fibres}\}, \\ \text{forest observables} \implies \{a, \sigma(m), \tau(n)\} \implies \text{iff signature and collisions.} \end{array}$$

Thus the signature theorem is not obtained merely from equal depth polynomials; it uses the full component structure and total vertex count.

## 2. FROM THE ONE-STEP RULE TO THE ENTIRE FOREST

For completeness, we recall why (1) and (2) hold. A subgroup contained in  $\langle r \rangle$  is a unique  $R_d$ . Otherwise its rotation part is  $R_d$ , and any reflection  $r^j s$  in it generates the unique nonrotation coset, giving  $H_{d, j}$  with  $j \bmod d$ . This also proves the count  $\tau(n) + \sigma(n)$ .

Every  $R_d$  is normal in  $G_n$ . Conjugation by  $r^u$  sends  $r^j s$  to  $r^{j+2u} s$ , so it normalizes  $H_{d, j}$  exactly when  $d \mid 2u$ . A general normalizing reflection  $r^u s$  satisfies the corresponding condition  $d \mid 2(u - j)$ . Therefore all normalizers together form

$$\left\langle r^{d/\gcd(d, 2)}, r^j s \right\rangle,$$

which is precisely (2). This derivation makes the coordinate convention explicit, but the result itself is part of the credited input.

Write  $d = 2^k e$  with  $e$  odd. Iterating (2) removes one factor of two per step until none remains, and reduces  $j$  modulo the new step. This proves (3). For a fixed pair  $(e, j_0)$ ,  $e \mid m$  and  $0 \leq j_0 < e$ , the states at level  $k$  are

$$(10) \quad H_{2^k e, j_0 + \ell e}, \quad 0 \leq \ell < 2^k.$$

Each non-leaf state has exactly two lifts at the next level. Summing the number of roots over odd divisors gives  $\sum_{e \mid m} e = \sigma(m)$ , so these are  $\sigma(m)$  full binary trees of height  $a$ . Separately, all  $\tau(n)$  rotations map directly to  $H_{1, 0}$ . The claimed depths and (4) follow. Every orbit reaches a root, hence every recurrent state is fixed.

## 3. ALL-TIME IMAGES AND TARGET FIBRES

At time  $t \geq 1$ , a level- $\ell$  dihedral state can be reached exactly when  $0 \leq \ell \leq \max(a - t, 0)$ . Each of the  $\sigma(m)$  trees then contributes

$$1 + 2 + \dots + 2^{\max(a-t, 0)} = 2^{\max(a-t, 0)+1} - 1$$

image vertices, proving (5). Rotations have no preimages because every normalizer contains the subgroup it normalizes, whereas a proper rotation target omits all reflections; this is also immediate from (2).

Let  $H_{2^k e, j}$  be a carrier target with  $1 \leq k \leq a$ ,  $e \mid m$ , and  $0 \leq j < 2^k e$ . A time- $t$  source, for  $t \geq 1$ , must occur at level  $k + t$ . When that level exists, the residue  $j$  has exactly  $2^t$  lifts modulo  $2^{k+t} e$ ; otherwise

there is no source. This proves (6). For a root, all levels  $0, \dots, \min(t, a)$  have already arrived, giving the geometric sum  $2^{\min(t, a)+1} - 1$ . In addition, every rotation has arrived at every positive time, but only at the distinguished root. This gives (7).

These formulas also close source mass without appeal to graph intuition. Indeed, summing all positive-level fibres, all root-tree fibres, and the  $\tau(n)$  distinguished additions gives

$$\sigma(n) + \tau(n) = |\text{Sub}(G_n)|.$$

Targets absent from (5) have fibre zero. Thus (6)–(7) are an every-target atlas rather than only a distribution of indegrees.

As a concrete ledger, take  $n = 12 = 2^2 \cdot 3$ . Here  $a = 2$ ,  $\sigma(m) = 4$ ,  $\tau(n) = 6$ , and

$$(11) \quad |\text{Sub}(G_{12})| = \sigma(12) + \tau(12) = 34, \quad \mathcal{D}_{12}(z) = 4 + 14z + 16z^2.$$

At time one the image has  $4(2^2 - 1) = 12$  states. Each of the eight level-one targets has two sources; each ordinary root has three sources; and the distinguished root has  $3 + 6 = 9$ . Thus

$$8 \cdot 2 + 3 \cdot 3 + 9 = 34.$$

At time two only the four roots remain. The three ordinary roots have seven sources each and the distinguished root has  $7 + 6 = 13$ , giving  $3 \cdot 7 + 13 = 34$ . This box displays both mechanisms in (7): binary-tree mass grows geometrically, while the rotation contribution is already saturated at time one.

Because every cycle is a fixed root, the finite-map zeta function follows as

$$(12) \quad \zeta_{\mathcal{N}_n}(z) = \exp \left( \sum_{t \geq 1} \frac{\sigma(m)}{t} z^t \right) = (1 - z)^{-\sigma(m)}.$$

This is recorded for completeness; once the fixed-root census is known, it is generic cycle bookkeeping and receives no standalone credit.

#### 4. EXACT INVERSE SIGNATURE

Let

$$S(n) = (a, \sigma(m), \tau(n)), \quad n = 2^a m, \quad m \text{ odd}.$$

If  $S(n) = S(q)$ , match the distinguished roots, match all other roots arbitrarily, preserve the binary positions in (10), and match the  $\tau(n)$  extra rotation leaves. This constructs a bijection commuting with the two maps, so equal signatures are sufficient.

Conversely, the unlabelled graph recovers  $\sigma(m)$  as its number of fixed vertices. Let  $L$  be its maximum tail. If  $L \geq 2$ , then  $a = L$ . The only ambiguity is  $L = 1$ , where  $a \in \{0, 1\}$ . Because  $n \geq 3$ , in either case the relevant odd part exceeds one and there is more than one fixed root. If  $a = 0$ , only the distinguished root has nonfixed predecessors. If  $a = 1$ , every root has two level-one dihedral predecessors. Hence the graph distinguishes the two cases and recovers  $a$ .

Now the complete population of dihedral vertices is

$$\sigma(m)(1 + 2 + \dots + 2^a) = \sigma(m)(2^{a+1} - 1).$$

The remaining vertices are exactly the rotations, so the graph recovers

$$(13) \quad \tau(n) = |V| - \sigma(m)(2^{a+1} - 1).$$

This total-vertex formula is essential: a shortcut comparing the distinguished root with other roots would not cover the one-root case  $m = 1$ . Equation (13) covers it without exception and completes necessity in (8).

The 33/35 collision. Both parameters are odd and

$$\sigma(33) = 48 = \sigma(35), \quad \tau(33) = 4 = \tau(35).$$

All 48 dihedral vertices are fixed. Match  $H_{1,0}$  with  $H_{1,0}$ , match the remaining  $(d, j)$  pairs lexicographically, and match the four rotation leaves by

$$(14) \quad R_1 \mapsto R_1, \quad R_3 \mapsto R_5, \quad R_{11} \mapsto R_7, \quad R_{33} \mapsto R_{35}.$$

Every rotation maps to the distinguished root, so this 52-state bijection commutes with the updates. Multiplication by  $2^b$  preserves  $a = b$ , the odd-part divisor sum 48, and  $\tau(2^b 33) = 4(b + 1) = \tau(2^b 35)$ ; (8) gives every lifted collision.

Sharpness and boundary examples. The third signature coordinate cannot be dropped. For example,

$$v_2(15) = v_2(23) = 0, \quad \sigma(15) = \sigma(23) = 24,$$

but  $\tau(15) = 4$  and  $\tau(23) = 2$ ; the graphs have 28 and 26 vertices. At the other extreme, (14) shows that even the complete graph does not identify the ambient group order. We use the literal subgroup set, not conjugacy classes of subgroups: quotienting by conjugacy merges residue positions and changes the binary fibres. Finally, the restriction  $n \geq 3$  keeps the standard nondegenerate dihedral convention; no statement here uses  $n = 1, 2$  to resolve the  $a = 0/1$  case.

## 5. OWNER SUBTRACTION, COLLISION FIREWALL, AND EXACT AUDIT

TABLE 1. Owner subtraction and internal system firewall.

| Source or neighbour             | Material receiving no separation credit  | Literal separation retained here                                                                                                            |
|---------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Cavior [1], Conrad [2]          | Complete subgroup coordinates and counts | None until the coordinates are iterated as one global map                                                                                   |
| Frenkel [3], Shelash et al. [4] | Odd/even one-step normalizer and halving | Full forest, all-time target fibres, iff signature, and order collisions                                                                    |
| P91 / P135                      | Finite-group or centralizer vocabulary   | No relation shift, spectrum, wreath product, or partition orbit; the carrier here is every subgroup and the arrow is its ambient normalizer |
| P142 reserve                    | Divisors and two-adic halving            | Literal subgroup normalizers and an unlabelled-graph inverse theorem, not a divisor-only gcd map                                            |
| P153                            | Finite noninvertible branches            | Binary forests into fixed roots, not factorial arms into a translating field cycle                                                          |

Related two-adic normalizer-length calculations, such as the Wielandt subgroup setting of Shelash and Ashrafi [5], are also treated as structural adjacency, not as support for a contribution claim. Formula searches used dihedral-subgroup coordinates, normalizer towers, hypernormalizers, parity halving, functional graphs, and the 33/35 arithmetic signature. The bounded search found the direct one-step owners above, but no source stating the entire iterated graph–fibre–signature conjunction. That bounded result is not an ownership certificate.

Deterministic replay. The accompanying `verify.py` uses no randomness or external package. For 44 values of  $n$ , it materializes every displayed subgroup as an element set, computes every normalizer by conjugating every subgroup element with every ambient group element, and only then compares with (2). It next checks depths, fixed roots, images, and every target fibre through three steps beyond stabilization. Four equal-signature pairs, beginning with 33/35, receive explicit commuting bijections. The frozen `CANONICAL.txt` ends with

```
PROFILE_SHA256 6eed12ce0c63f2d20f734ac1fa67634ce445140372dfc53e779a389de023b782
TOTAL boxes=44 iso_pairs=4 assertions=29590
VERDICT PASS_EXACT_REPLAY
```

An independent pressure scan recorded no signature mismatch for  $3 \leq n \leq 500$ , but bounded enumeration is only a regression and falsification control. Subgroup completeness and both directions of (8) are supplied by the proofs above.

Reproducibility and release state. The repository includes the exact script, canonical transcript, claim–evidence matrix, source-verification ledger, controls, and deterministic build instructions. The manuscript is anonymous and uses no human-subject data or randomized inference. External circulation remains `HOLD_EXTERNAL`.

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